

RGB Drone Detection Baseline: YOLOv2 vs YOLO26n

What was done

Two object detectors were trained independently on the same ordinary RGB video frames, as a baseline against which future event-based models can be compared. Source data is the Halmstad Drone-detection-dataset (CC0, Svanstrom et al.): 114 drone-labelled clips at 640x512 / 30 fps, 36,316 annotated boxes.

Frames were letterboxed to 224x224 (scale 0.3500, 22.4 px padding top and bottom), giving 35,496 samples. Splits are by clip, never by frame, so no clip contributes to more than one split: 90 train / 12 validation / 12 test, stratified by median target size. The same geometry and splits were used for both models and for the earlier event-based work, so all results are directly comparable.

Targets are small, and this dominates every result below. Median equivalent square is 10.4 px (a third of a grid cell); 90.8% of boxes are under 16 px, 15.9% under 8 px.

Models

	YOLOv2 (akida_models)	YOLO26n (Ultralytics)
backbone	AkidaNet, alpha 0.5	C3k2 / C2PSA
parameters	3,568,590	2,375,031
pretraining	none, trained from scratch	COCO
anchors	5, IoU k-means, mean IoU 0.818	anchor-free
output	single 7x7 grid	three detection scales
augmentation	none	mosaic, flips, HSV, scale
training	30 epochs, lr 1e-3, batch 64	50 epochs, MuSGD auto-tuned, batch 64

Results

	YOLOv2 val	YOLOv2 test	YOLO26n val	YOLO26n test
AP@0.5	0.431	0.288	0.833	0.779
precision	0.541	0.471	0.849	0.822
recall	0.549	0.459	0.839	0.750
mAP@0.5:0.95	-	-	0.372	0.269

YOLOv2 recall by target size (validation): 0.326 under 8 px, 0.617 at 8-12 px, 0.730 at 12-16 px, 0.341 above 16 px. YOLO26n inference is 1.1 ms per image on a T4, about 2 ms end to end, roughly 16x faster than the 30 fps source video.

Reading the numbers

YOLO26n roughly doubles YOLOv2 on identical data, with fewer parameters. The gap is not architecture alone: COCO pretraining, three detection scales instead of one coarse 7x7 grid, and real augmentation all contribute, and YOLOv2 had none of them. An architecture-only comparison would require pretraining and augmenting the YOLOv2 side too.

YOLOv2 generalises poorly. It loses a third of its performance from validation to test (0.431 to 0.288), while YOLO26n loses 7%. Training recall reached 0.98 against validation recall 0.63, which is memorisation: consecutive video frames are near-duplicates, so 28,045 samples are far fewer independent examples than the count suggests, and nothing countered it.

The YOLOv2 anchor set is a real flaw. Recall above 16 px (0.341) is worse than at 12-16 px (0.730), which is backwards. The largest anchor covers only 1.5% of training boxes, so that size was barely learned. YOLO26n is anchor-free and does not have this failure.

Low mAP@0.5:0.95 is expected, not a defect. Passing IoU 0.9 on a 10 px box needs sub-pixel edge accuracy the ground-truth labels themselves do not have.

Against the event-based pipeline

The earlier event-based YOLOv2 (20 ms windows, messy preset, 1 channel) scored AP 0.428 on validation against 0.431 for the same architecture on RGB, on identical splits and geometry. **At this model size the event representation loses essentially nothing relative to ordinary video**, which is the useful finding for the event-based direction.

Limitations

Frames were pre-letterboxed to 224x224 before Ultralytics applied its own resizing, so padding becomes trained content that does not appear at inference. This was deliberate, to keep pixels identical between the two models; it applies equally to both and so does not explain the gap, but a YOLO26n run on raw 640x512 frames would likely score higher. Test and validation medians match but their tails do not: sub-8px boxes are 25.0% of validation and 4.6% of test, so the splits are not equally hard. One seed per model.